

Problem Set 6: Causal Inference, RCTs & Structural +
Experimental Integration
Microeconomic Theory for Experimentalists & Applied Microeconomists

Due: [date — before Lecture 1 of Module 7]

Ground rules: collaboration on ideas is encouraged; write-ups are individual. You may use AI tools for parts flagged with $\langle AI \rangle$ and must attach the transcript. Show all calculations.

Weights: Part A — 30 points. Part B — 45 points. Part C — 25 points.

Part A: Theory — Guided Calculations (30 points)

Question 1 (Potential outcomes and selection bias, by hand — 15 points). *Six workers face a voluntary training program. An omniscient observer sees both potential outcomes (monthly earnings, in hundreds):*

Person	Y_0 (untrained)	Y_1 (trained)	Enrolls voluntarily?
1	30	50	yes
2	50	60	yes
3	70	70	no
4	40	60	yes
5	60	70	yes
6	80	80	no

- (a) **(3 pts)** Compute each person's treatment effect and the average treatment effect (ATE). Which people does the program help, and which does it leave unchanged?
- (b) **(5 pts)** Under voluntary enrollment, the analyst observes Y_1 for enrollees and Y_0 for non-enrollees. Compute the naive comparison (mean observed outcome of enrollees minus mean observed outcome of non-enrollees) and note its sign.
- (c) **(5 pts)** Decompose the naive comparison into the average effect on the treated (ATT) plus selection bias, computing each term numerically and verifying they sum to your answer in (b). Hint: selection bias compares the enrollees' and non-enrollees' UNTREATED outcomes. One sentence: whose rational behavior created the bias?
- (d) **(2 pts)** Now enrollment is assigned by coin flip. State what the difference in group means estimates, and name two questions about this program that even a perfectly executed RCT does NOT answer.

Question 2 (Power by simulation — 15 points). You are planning a correspondence study. Piloting suggests callback rates of 9.7% for the control name group and 6.5% for the treatment name group (the Bertrand–Mullainathan magnitudes). A classmate runs the Lab 6 power simulation (2,000 simulated experiments per sample size, two-sided test at the 5% level) and reports:

<i>n</i> per arm	500	1,000	1,500	2,000
Estimated power	0.46	0.75	0.90	0.96

- (a) **(4 pts)** In words a first-year student would understand: what does “power = 0.75 at $n = 1,000$ ” mean operationally, and why does power rise with n , rise with the true gap, and fall with outcome variance?
- (b) **(4 pts)** Recommend a sample size for 80% power, justifying your number from the table (interpolation is fine; state your reasoning). What total number of resumes does your recommendation imply, and why might you inflate it further before fielding the study?
- (c) **(4 pts)** The budget caps the study at $n = 2,000$ per arm. The simulation reports that the standard error of the estimated gap at that size is about 0.0086. Compute the minimum detectable effect at 80% power (use $MDE \approx 2.8 \times SE$) and interpret it in one sentence: which true gaps would this study likely miss?
- (d) **(3 pts)** Your advisor says: “skip the simulation, use the standard formula.” Give two concrete design features of a real correspondence study that the simulation can price and the textbook two-proportion formula cannot, and one sentence on why making assumptions explicit is a feature, not a cost.

Part B: Experimental Design (45 points)

Question 3 (Separating taste-based from statistical discrimination). Bertrand & Mullainathan (2004) established the callback gap. Design a correspondence study whose goal is the harder question: is the gap driven by taste-based discrimination (employers dislike hiring the group, information cannot help) or statistical discrimination (group membership proxies for unobserved productivity, so individuating information should make the proxy redundant)? Specify:

- (a) **(6 pts)** The market: occupations, cities, how you find vacancies, and how many applications per vacancy you can send without detection — and why your occupation choice matters for the taste-vs-statistical question.
- (b) **(12 pts)** Treatments. Your design must cross the name manipulation with at least one information-richness manipulation (e.g. verifiable certifications, references, licenses on the resume) and state the predicted callback pattern under each discrimination model — including what the interaction term tells you that the two main effects cannot.
- (c) **(9 pts)** Mechanics and ethics: how resumes are constructed and matched, how names are assigned, how callbacks are captured and coded, and the two main ethical costs of audit designs (employer deception and wasted screening time) with the standard mitigations an IRB will expect.

- (d) **(9 pts)** *Hypotheses and power: anchor on BEM — treat 9.7%/6.5% as the baseline-arm callback rates — and your Part A(2) simulation logic; state the per-cell and total application counts your design implies once the information arms are added, and identify which comparison (main effect or interaction) drives the sample size.*
- (e) **(9 pts)** *The biggest confound in YOUR design and your answer to it. (Candidates: names signal socioeconomic status, not only race; information manipulations change perceived overqualification; detection risk correlates with arm.)*

Part C: Silicon Pilot Interpretation <AI> (25 points)

Question 4 (Whose power calculation is it?). You run `module6-simulation-lab.ipynb`. Your silicon hiring managers produce a callback gap of 2.1 percentage points between name groups, with very tight variance across runs. Your lab partner proposes: “feed the silicon effect size and variance into our power calculator — it says we only need 800 resumes per arm, so we can cut the human study’s budget by half.”

- (a) **(8 pts)** *Give two distinct reasons the silicon effect size is not a valid stand-in for the human effect size (at least one specific to how LLMs are trained and aligned), and one reason the silicon variance is even less trustworthy than the silicon mean.*
- (b) **(9 pts)** *Write the power-calculation protocol you WOULD defend to an advisor: state where each input (effect size, base rate, variance/ICC, attrition) should come from, what role — if any — the silicon pilot legitimately plays in the protocol, and the sensitivity analysis you would attach.*
- (c) **(8 pts)** *Now the twist: argue that your silicon result is not merely a failed pilot but a publishable object in its own right — LLM resume screeners are deployed technology. State the research question your silicon audit actually answers, one robustness exercise it would need to be credible (the exercise may span multiple models and prompt framings), and the one-sentence scope limitation you would put in its abstract.*

Attach your AI transcripts and, if you ran code, the results CSV.